

Network-Validated Distributed ADMM Coordination of Cold-Climate Electric Heating under Communication Failures

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I. INTRODUCTION

THE electrification of heating in cold-climate regions concentrates distribution-system stress into a few winter hours. In the province of Québec, where electric space heating is the dominant heating technology, the system peak is driven almost entirely by simultaneous resistive and heat-pump heating on the coldest mornings and evenings. As distributed energy resources (DERs) proliferate and heating loads grow, these synchronized peaks threaten thermal limits on feeder trunk conductors and depress voltages far from the substation. Because the flexibility lives behind thousands of meters and not at a single controllable node, centralized dispatch scales poorly; decentralized coordination, in which each home solves a small local problem and exchanges only an aggregate signal, is the natural fit.

The alternating direction method of multipliers (ADMM) [1] is a workhorse for this setting: its sharing form lets a coordinator flatten a common resource (here, total feeder load) while each agent retains private control of its own heating schedule. ADMM-based demand response is well studied, yet two gaps undermine its real-world credibility.

The first gap is fragility to communication failures. Multi-agent coordination implicitly assumes every agent reports on every iteration; when a fraction of agents fall silent, the aggregate the coordinator optimizes is no longer the aggregate that materializes on the wire. How badly the coordinated outcome degrades — and how an operator might recover it by forecasting the missing contributions — is rarely quantified end to end.

The second gap is physical validity. The literature typically stops at the coordinated aggregate kilowatt signal and reports peak or peak-to-average reductions. It almost never asks the operational question that matters: is the coordinated aggregate actually feasible on the feeder? A flatter aggregate

is worthless if it still overloads a trunk line or sags a remote bus below limits.

This paper closes the loop between coordination and feeder physics. We adapt the sharing-ADMM-plus-imputation methodology from a prior multi-agent thesis and extend it with an AC power-flow validation stage, turning a purely signal-level study into a network-validated one. Our contributions are:

- A sharing-ADMM coordinator for deferrable electric heating that flattens the feeder aggregate subject to per-home comfort bounds and a daily-energy conservation constraint (load is shifted, never curtailed).
- A LightGBM imputer that reconstructs the heating of non-responsive agents from ambient temperature, time-of-day, and nameplate features, letting the coordinator stay accurate as the non-responsive fraction grows.
- An AC power-flow validation stage on the IEEE 33-bus feeder in pandapower that distributes the coordinated aggregate across load buses, calibrates trunk ampacity to a realistic uncoordinated overload, and records worst-of-day minimum voltage and maximum line loading — closing the loop on physical feasibility.
- A fully reproducible, governed study driving the entire pipeline from a declarative configuration, with every reported number traceable to a recorded artifact.

The central finding is that coordination not only flattens the aggregate (a 12.7% peak cut) but clears a 115% thermal overload to 99.7%, and that this cleared state holds under communication failure up to a non-responsive fraction of 0.6 before re-violating at 0.8.

The remainder of the paper is organized as follows. Section IV is preceded by the related work and the system model; Section II surveys ADMM-based demand response, multi-agent resilience, telemetry imputation, and distribution

power flow, and states our delta against each. Section III formalizes the agents, the feeder, and the communication-failure model. Section IV develops the sharing-ADMM updates, the LightGBM imputer, and the power-flow validation loop. Section V details the case-study configuration, Section VI reports the aggregate, robustness, and network-feasibility results, and Section VII concludes.

II. RELATED WORK

A. ADMM AND SHARING FOR DEMAND RESPONSE

ADMM [1] decomposes a coupled optimization into local subproblems linked by a consensus or sharing constraint, and has become a standard tool for distributed energy coordination. The sharing form (Boyd et al., Section 7.3) is particularly suited to demand flattening, where many agents contribute to a single shared resource. Asynchronous and energy-grid-specific variants extend it to microgrids and optimal power flow [2], [3]. These works establish that ADMM converges to a flattened aggregate; they do not, in general, check that the aggregate is feasible on the underlying network.

B. MULTI-AGENT RESILIENCE TO NON-RESPONSIVENESS

A separate line of work studies how multi-agent coordination degrades when agents lose communication. Brown et al. show that even a single lost link can drive game-theoretic multi-agent systems to arbitrarily poor states [4], and consensus-control surveys catalog fault-tolerant mechanisms [5]. In the distributed-optimization setting, communication delay and loss have been shown to slow or destabilize ADMM-based optimal power flow [6]. This literature characterizes when coordination breaks but rarely couples the failure model to a concrete recovery mechanism whose error is then propagated into feeder physics.

C. IMPUTATION OF MISSING TELEMETRY

When agents fall silent, the coordinator must estimate their contribution. Machine-learning imputation of missing time-series telemetry, and its downstream effect on forecasting, is an active area [7], with cooperative network estimators able to fill missing data across agents [8]. Gradient-boosted trees such as LightGBM [9] are strong tabular regressors and are well matched to imputing a single agent's heating from weather and calendar features.

D. DISTRIBUTION POWER FLOW

Physical validation requires a feeder model and a solver. The IEEE 33-bus radial test feeder [10] is the canonical distribution benchmark, and pandapower [11] provides convenient AC Newton-Raphson power flow over it. These tools are mature; what is missing is their systematic use as the acceptance test for a coordination scheme.

E. THE DELTA

Prior work treats coordination quality and network feasibility as separate concerns: ADMM papers report aggregate metrics, network papers assume a given load. We connect them. By feeding the coordinated (and imputation-perturbed) aggregate into an AC power flow and reading back thermal and voltage violations, we measure coordination by the metric an operator cares about: does the feeder stay within limits? To our knowledge this closed loop — ADMM, imputation, and power-flow validation under a swept communication-failure fraction — has not been reported for cold-climate electric heating.

III. SYSTEM MODEL

A. AGENTS AND DEFERRABLE HEATING

We model $N = 30$ representative cold-climate electric-heating homes over a single synthetic Trois-Rivières cold day at 15-minute resolution ($T = 96$ time steps). Each home i has a thermostat-driven baseline heating profile $h_{i,t}$ produced by a behavioral model conditioned on the ambient temperature, plus a fixed (non-deferrable) background load. The coldest-day minimum temperature is -21.9°C (daily mean -18.3°C), which sets a demanding heating duty.

Heating is treated as deferrable within a comfort band of $\pm 50\%$ of the baseline ($\alpha = 0.5$): the coordinator may shift when a home heats but must conserve the home's total daily heating energy, so demand is shifted rather than curtailed. With $x_{i,t}$ the coordinated heating power of home i at step t , each agent is constrained by

$$(1 - \alpha) h_{i,t} \leq x_{i,t} \leq (1 + \alpha) h_{i,t}, \quad \sum_t x_{i,t} = \sum_t h_{i,t}. \quad (1)$$

The box bound preserves thermal comfort at every step, and the equality plane preserves daily heating energy. Background load is held fixed and is not subject to (1).

B. THE FEEDER

Physical feasibility is assessed on the IEEE 33-bus radial test feeder [10] as implemented in pandapower [11] (the case33bw case): 33 buses, 32 lines, 32 distributed loads, a single external-grid substation at the head, and a nominal voltage of 12.66 kV. The coordinated aggregate is allocated to the 32 load buses in proportion to their native load shares and scaled so that the uncoordinated winter peak reaches the feeder. The trunk conductor ampacity is calibrated so that the uncoordinated peak loads the worst line to 115% — a realistic overload that motivates demand response in the first place. All loads use a power factor of 0.95.

C. COMMUNICATION-FAILURE MODEL

At coordination time a fraction ρ of the agents become non-responsive and report nothing to the coordinator. We sweep $\rho \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$, corresponding to 0, 6, 12, 18, and 24 silent homes out of 30. For each non-responsive

Algorithm 1: Sharing-ADMM heating coordination

Input: baselines $\{h_i\}$, background B , target c ,
weights λ, μ , penalty ρ_{ADMM}
Output: coordinated schedules $\{x_i\}$
 initialize $x_i \leftarrow h_i, z \leftarrow 0, u \leftarrow 0$;
repeat
 foreach home i (in parallel) **do**
 proximal step on $\frac{\lambda}{2}\|x_i - h_i\|^2$ plus penalty
 toward $z - u$;
 project x_i onto the box-and-energy set of (1);
 $z \leftarrow$ closed-form average enforcing the flat
 aggregate target;
 $u \leftarrow u + (\sum_i x_i + B) - z$;
until aggregate residual converged;
return $\{x_i\}$;

agent the coordinator substitutes an imputed heating profile $\hat{x}_{i,t}$ produced by a trained regressor (Section IV) and proceeds as if it were the true contribution. The imputation error therefore propagates first into the coordinated aggregate and ultimately into the feeder power flow, which is exactly the propagation path this study quantifies.

IV. METHODOLOGY

A. SHARING-ADMM COORDINATION

Let $x_i \in \mathbb{R}^T$ be the heating schedule of home i , h_i its baseline, and $B \in \mathbb{R}^T$ the fixed background aggregate. The coordinator flattens the total feeder load toward a constant target c while respecting each home's comfort. We solve the sharing problem

$$\min_{\{x_i\}} \sum_{i=1}^N \frac{\lambda}{2} \|x_i - h_i\|_2^2 + \frac{\mu}{2} \left\| \sum_{i=1}^N x_i + B - c \mathbf{1} \right\|_2^2 \quad (2)$$

subject to (1) for every i , where c is the constant level that conserves total daily energy, $\lambda = 0.15$ weights comfort deviation, $\mu = 1.0$ weights flattening, and $\mathbf{1}$ is the all-ones vector. The first term keeps each home close to its comfortable baseline; the second penalizes any departure of the total load from a flat profile.

Following the sharing form of ADMM [1], we introduce an auxiliary aggregate variable z and a scaled dual u with penalty $\rho_{\text{ADMM}} = 1.0$, and iterate three steps. The **x -update** solves each home's local problem in parallel: a comfort-plus-penalty proximal step followed by projection of the result onto the box-and-energy-plane feasible set of (1). The **z -update** is the closed-form average that drives the shared aggregate toward the flat target. The **u -update** is the standard scaled dual ascent on the sharing residual. Algorithm 1 summarizes one full sweep.

The x -update projection is what makes the scheme physically meaningful: it guarantees that every iterate respects comfort and conserves daily energy, so the coordinated

schedule is feasible for the homes by construction. The number of iterations to convergence is recorded as a tractability metric.

B. IMPUTATION OF NON-RESPONSIVE AGENTS

When agent i is non-responsive its baseline h_i is unavailable to the coordinator, so we replace it with a learned estimate $\hat{h}_i$. We train a single LightGBM gradient-boosted-tree regressor [9] that maps a small feature vector — ambient temperature, the sine and cosine of the hour of day, and the agent's nameplate heating level — to instantaneous heating power. The model is trained on the responsive population and applied step-by-step to each silent agent, yielding $\hat{x}_{i,t}$ that the coordinator treats as the agent's contribution. Generalization to unseen homes is assessed by a 5-fold leave-agents-out cross-validation, which holds out entire agents rather than random samples and therefore measures the realistic case of imputing a home the model never saw.

C. POWER-FLOW VALIDATION LOOP

The novel stage closes the loop on physics. For each scenario (uncoordinated, ideal, and each ρ) we take the resulting aggregate, distribute it across the 32 IEEE-33 load buses by their native load shares, and scale so that the uncoordinated peak reaches the calibrated feeder. The trunk ampacity is fixed so the uncoordinated peak loads the worst line to 115%. We then run an AC Newton-Raphson power flow at every one of the 96 time steps (672 solves across all scenarios, all converged) and record, per scenario, the worst-of-day minimum bus voltage and the worst-of-day maximum line loading. A scenario is flagged as violating if any line exceeds 100% loading at any step. This turns an abstract kilowatt signal into a binary feeder verdict and a pair of continuous severity measures.

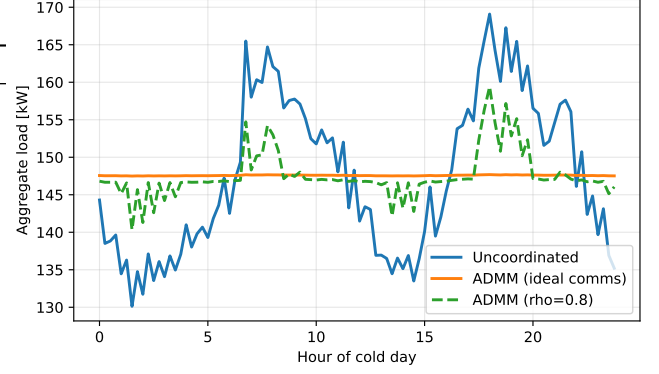
Three modeling choices deserve comment. Load allocation by native share keeps the spatial signature of the IEEE-33 case intact, so the worst line is the one the benchmark already stresses (the trunk leaving the substation) rather than an artifact of our placement. Ampacity calibration to 115% is what makes the experiment decision-relevant: a feeder that is comfortably within limits in the uncoordinated case has nothing to gain from demand response, so we set the binding constraint exactly where coordination must act to clear it. Solving every step rather than only the aggregate-peak step matters because imputation error can shift the worst instant: the uncoordinated worst is the evening peak, but under heavy imputation the binding step can migrate, and an aggregate-peak-only check would miss it. Reporting the worst-of-day envelope across all 96 steps therefore gives a conservative, instant-agnostic verdict.

V. CASE STUDY SETUP

The study runs on a single synthetic Trois-Rivières cold day with a minimum ambient temperature of -21.9°C , discretized at 15-minute resolution into $T = 96$ steps. We coordinate $N = 30$ electric-heating homes with comfort

TABLE 1. Principal case-study parameters.

Parameter	Value
Homes N	30
Horizon / resolution	96 steps / 15 min
Coldest-day min. temperature	-21.9°C
Comfort band α	0.5
Comfort weight λ	0.15
Flattening weight μ	1.0
ADMM penalty ρ_{ADMM}	1.0
Non-responsive fraction ρ	$\{0.0, 0.2, 0.4, 0.6, 0.8\}$
TOU off / mid / peak (CAD/kWh)	0.078 / 0.128 / 0.158
Feeder	IEEE 33-bus, 12.66 kV
Power factor	0.95
Uncoordinated worst-line loading	115% (calibrated)
Power flow	AC Newton-Raphson
Software	pandapower 3.4.0, scikit-learn 1.6.1, LightGBM 4.6.0

**FIGURE 1.** Aggregate cold-day load: uncoordinated double-peak versus the flat ADMM ideal profile (≈ 147.7 kW) and the $\rho = 0.8$ imputed case with residual peaks.

band $\alpha = 0.5$, comfort weight $\lambda = 0.15$, flattening weight $\mu = 1.0$, and ADMM penalty $\rho_{\text{ADMM}} = 1.0$. The communication-failure fraction is swept over $\rho \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$.

Energy cost is evaluated against a time-of-use tariff with an off-peak rate of 0.078, a mid-peak rate of 0.128, and an evening-peak rate of 0.158 CAD/kWh. Physical feasibility uses the IEEE 33-bus feeder with all loads at a power factor of 0.95, with the aggregate scaled so the uncoordinated winter peak reaches the feeder and trunk ampacity calibrated to a 115% uncoordinated overload. The software stack is pandapower 3.4.0 for power flow, scikit-learn 1.6.1 for the cross-validation harness, and LightGBM 4.6.0 for the imputer. Every input, intermediate, and reported figure is emitted as a governed artifact so the run is reproducible from its declarative configuration. Table 1 collects the principal parameters.

VI. RESULTS

A. AGGREGATE COORDINATION

Figure 1 shows the cold-day aggregate load. The uncoordinated profile exhibits the classic double peak — a morning and an evening heating surge — whereas the ideal ADMM coordination flattens it to a nearly constant ≈ 147.7 kW. The $\rho = 0.8$ trace, in which most agents are imputed, recovers much of the flattening but leaves residual peaks where imputation error accumulates.

Table 2 quantifies the aggregate key performance indicators. Ideal coordination cuts the peak from 169.1 to 147.7 kW, a 12.7% reduction, and drives the peak-to-average ratio (PAR) from 1.146 down to 1.001 — essentially a flat load. A PAR of 1.001 is the practical floor: it means the coordinated aggregate is constant to within a tenth of a percent across the entire day, so the flattening objective in (2) has been met almost exactly. Energy cost falls slightly, from 371.0 to 364.8 CAD, because the daily-energy constraint conserves total heating; the saving comes from shifting load out of the evening-peak TOU window (0.158 CAD/kWh) toward off-peak hours (0.078 CAD/kWh) rather than from using less energy. The modest size of the cost reduction

TABLE 2. Aggregate KPIs by scenario.

Scenario	Peak (kW)	PAR	Cost (CAD)	RMSE (kW)
Uncoordinated	169.1	1.1458	371.0	—
Coordinated ideal	147.7	1.0007	364.8	—
Imputed $\rho=0.0$	147.7	1.0007	364.8	0.000
Imputed $\rho=0.2$	147.7	1.0009	364.9	0.170
Imputed $\rho=0.4$	147.8	1.0012	364.9	0.242
Imputed $\rho=0.6$	148.1	1.0035	364.9	0.252
Imputed $\rho=0.8$	159.4	1.0802	366.2	0.234

is itself a useful result: with energy conserved, demand response of this kind buys capacity relief (a lower peak) far more than it buys energy savings, and the case for it must rest on the network benefit rather than the bill.

The imputed scenarios track the ideal closely up to $\rho = 0.6$ and only degrade materially at $\rho = 0.8$, where the peak rises to 159.4 kW and PAR to 1.080. The per-scenario imputation RMSE (last column) is non-monotone in ρ : it grows from 0.170 to 0.252 kW as the imputed population grows, then dips to 0.234 kW at $\rho = 0.8$. This is an averaging effect — with 24 of 30 agents imputed, individual errors partially cancel in the aggregate — and it explains why the aggregate-level damage at $\rho = 0.8$ comes less from raw imputation error than from the loss of the responsive agents whose true, flexible schedules were doing most of the flattening work.

B. IMPUTATION ACCURACY AND ROBUSTNESS

The LightGBM imputer is a competent forecaster of an unseen home's heating. Under 5-fold leave-agents-out cross-validation it achieves an RMSE of 0.294 kW and an MAE of 0.247 kW, against a heating standard deviation of 0.575 kW. At roughly 51% of the natural variability, the imputer is clearly better than predicting the mean and is accurate enough that the coordinated aggregate barely moves for small non-responsive fractions, as seen in the per-scenario RMSE column of Table 2.

Figure 2 plots peak and PAR against ρ . Both stay essentially flat through $\rho = 0.6$ and then turn sharply upward at

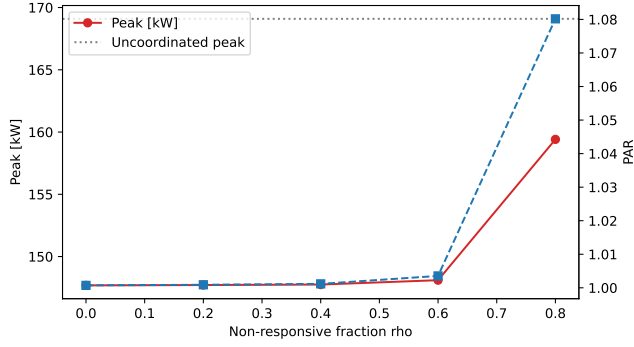


FIGURE 2. Aggregate peak and PAR versus the non-responsive fraction ρ .

$\rho = 0.8$, the signature of a robustness threshold rather than a gradual decay.

A secondary finding concerns tractability rather than outcome quality. The ADMM iteration count grows steeply with ρ : 8 iterations for the ideal case, then 14, 21, 150, and 280 for $\rho = 0.2, 0.4, 0.6, 0.8$ respectively (all runs converged). Communication failure thus degrades coordination tractability — the effort to reach consensus — well before it degrades the coordinated outcome, an early-warning signal an operator could monitor.

The shape of this curve is informative. From $\rho = 0.0$ to $\rho = 0.4$ the iteration count roughly doubles per step (8, 14, 21), an almost linear growth that the coordinator absorbs easily. Between $\rho = 0.4$ and $\rho = 0.6$ it jumps by a factor of seven (21 to 150), and by $\rho = 0.8$ it reaches 280 — 35 times the ideal effort. The knee of this curve sits at the same place as the feasibility knee in the network results below, near $\rho = 0.6$. The two are linked: as more agents are imputed, the aggregate the coordinator sees diverges from the aggregate that physically forms, the sharing residual becomes harder to drive to zero, and convergence slows. Iteration count is therefore a cheap, purely software-side proxy for the harder-to-observe physical-feasibility margin, because both respond to the same underlying quantity — the mismatch injected by imputation.

C. NETWORK VALIDATION

The closed-loop contribution is summarized in Figure 3 and Table 3. The uncoordinated aggregate loads the worst trunk line to 115% — a thermal violation — and drops the worst minimum voltage to 0.905 p.u. Ideal coordination clears the thermal overload: maximum line loading falls to 99.7% (no violation) and worst minimum voltage rises to 0.918 p.u. This is the core result: flattening the aggregate is not merely a statistical improvement, it restores feeder feasibility.

The cleared state is resilient to communication failure up to $\rho = 0.6$, where the worst line rides the 100.0% edge but does not violate. At $\rho = 0.8$ the accumulated imputation error re-introduces a 108% overload — the feeder re-violates. The story is therefore 115% $\rightarrow$ 99.7% $\rightarrow$ 108%: coordination clears the overload, the clearance survives substantial

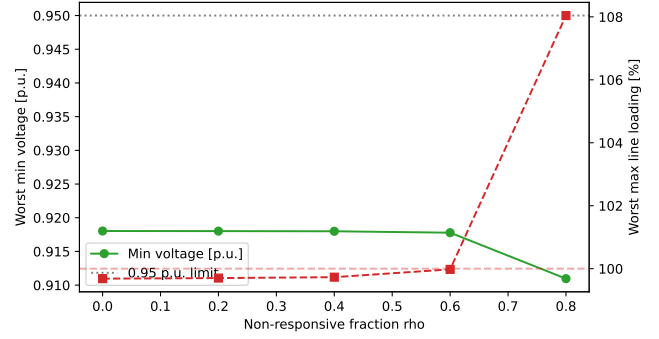


FIGURE 3. Worst minimum bus voltage (left axis) and worst maximum line loading (right axis) versus the non-responsive fraction ρ — the headline feasibility figure. Line loading clears below 100% through $\rho = 0.6$ and re-violates at $\rho = 0.8$.

TABLE 3. Network feasibility on IEEE-33 (worst of day).

Scenario	Peak (MW)	$V_{\min}$ (p.u.)	Loading (%)	Violation
Uncoordinated	4.60	0.905	115.0	yes
Coordinated ideal	4.02	0.918	99.7	no
Imputed $\rho=0.0$	4.02	0.918	99.7	no
Imputed $\rho=0.2$	4.02	0.918	99.7	no
Imputed $\rho=0.4$	4.02	0.918	99.7	no
Imputed $\rho=0.6$	4.03	0.918	100.0	no (edge)
Imputed $\rho=0.8$	4.34	0.911	108.0	yes

non-responsiveness, and only collapses once most agents must be imputed. This identifies a critical non-responsive fraction between 0.6 and 0.8 beyond which imputation can no longer protect the feeder.

a: An honest note on voltage.

We report voltage as a severity measure, not a compliance verdict. On the *case33bw* benchmark the worst minimum voltage never reaches the 0.95 p.u. ANSI limit in any scenario — the feeder is inherently stressed — so we do not claim voltage compliance. Coordination nonetheless reduces under-voltage depth (0.905 to 0.918 p.u.), which is a genuine improvement in feeder condition. The clean cleared/re-violated headline comes from line loading, where the 100% thermal limit gives an unambiguous binary verdict.

D. DISCUSSION

Three observations tie the results together. First, the aggregate, tractability, and feasibility curves all share the same knee near $\rho = 0.6$: the iteration count, the peak, and the worst line loading all stay benign through $\rho = 0.6$ and break together at $\rho = 0.8$. This suggests that a single observable quantity — the divergence between the imputed and the realized aggregate — governs all three, and that an operator monitoring ADMM convergence effort has an early proxy for impending feeder violation. Second, the network result is not a restatement of the aggregate result: a 12.7% peak cut is a statistical improvement, but clearing a 115% overload to 99.7% is an operational state change from infeasible to feasible, and the two need not coincide on a different

feeder or load allocation. Third, the failure at $\rho = 0.8$ is graceful rather than catastrophic — 108% loading, not a non-converging power flow — which indicates that the imputer keeps the system in a recoverable regime even when most agents are silent. Together these support the central claim that coordination quality should be measured by feeder feasibility, and that imputation extends the range of communication failure over which that feasibility holds.

VII. CONCLUSION

We presented a network-validated pipeline that coordinates cold-climate electric heating with sharing-ADMM, reconstructs non-responsive agents with a LightGBM imputer, and — the closing contribution — validates the coordinated aggregate on the IEEE 33-bus feeder with AC power flow. The closed loop turns abstract flattening metrics into a feeder verdict: ideal coordination cuts the aggregate peak by 12.7% and clears a 115% thermal line overload down to 99.7%, while lifting the worst minimum voltage from 0.905 to 0.918 p.u. The cleared state is resilient to communication failure up to a non-responsive fraction of 0.6 and re-violates at 0.8 (108% loading), establishing a critical- ρ threshold between the two. A secondary finding is that ADMM iteration count explodes with ρ (8 to 280) well before the outcome degrades, offering an early-warning signal.

The study has clear limitations. It covers a single cold day and a single feeder; the communication-failure model is a stochastic non-responsive fraction rather than a protocol-level simulation of delays, drops, and retransmissions; and the `case33bw` voltage profile is inherently stressed, so voltage is reported as a severity measure rather than a compliance test. Future work will extend the loop to multiple feeders and days, study the spatial placement of agents on the feeder (which determines how much each contributes to the binding constraint), and couple the coordinator to a flexibility market so that the cleared headroom can be priced and settled.

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